

Data Analyst Capstone Project Report

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OUTLINE



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EXECUTIVE SUMMARY



- **Programming languages:**
JavaScript remains highly prominent in both current usage and future preferences, while TypeScript, Go, and Rust gain visibility in the future outlook.
- **Database technologies:**
PostgreSQL leads both current usage and future preferences, while Redis shows a notable rise in the future outlook.
- **Technology platforms & frameworks:**
Cloud platforms are prominent in current and future preferences, with AWS, Microsoft Azure, and Google Cloud among the major desired platforms.
- **Job market:**
Job-posting data shows substantial variation in demand across locations, with Washington DC having the highest number of postings among the analysed locations.
- **Salary & demographics:**
Average annual salaries vary considerably by programming language, while the respondent population is concentrated in the 25–44 age range and Bachelor's degree holders form the largest education group.

INTRODUCTION



- **Purpose of the Report**
 - Analyse current and future technology trends.
 - Examine programming languages, databases, job postings, and salaries.
- **Target Audience**
 - Technology professionals, employers, and workforce/training planners.
- **Value of the Analysis**
 - Identifies current technology leaders and emerging trends.
 - Supports data-driven technology and skill-development decisions.

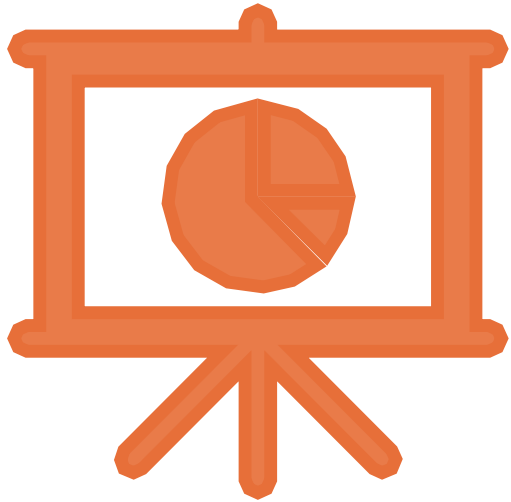


METHODOLOGY



- **Data Sources:** Stack Overflow Developer Survey and job-posting API data
- **Data Collection:** Survey datasets and API-based job-posting data were collected and organized for analysis
- **Data Preparation:** Relevant fields were selected, filtered, cleaned, and structured
- **Aggregation:** Respondent counts and Top 10 technology rankings were calculated
- **Job Analysis:** Job postings and salary data were organized by location and programming language
- **Visualization:** Charts and dashboards were created to compare current usage, future preferences, employment trends, and demographics

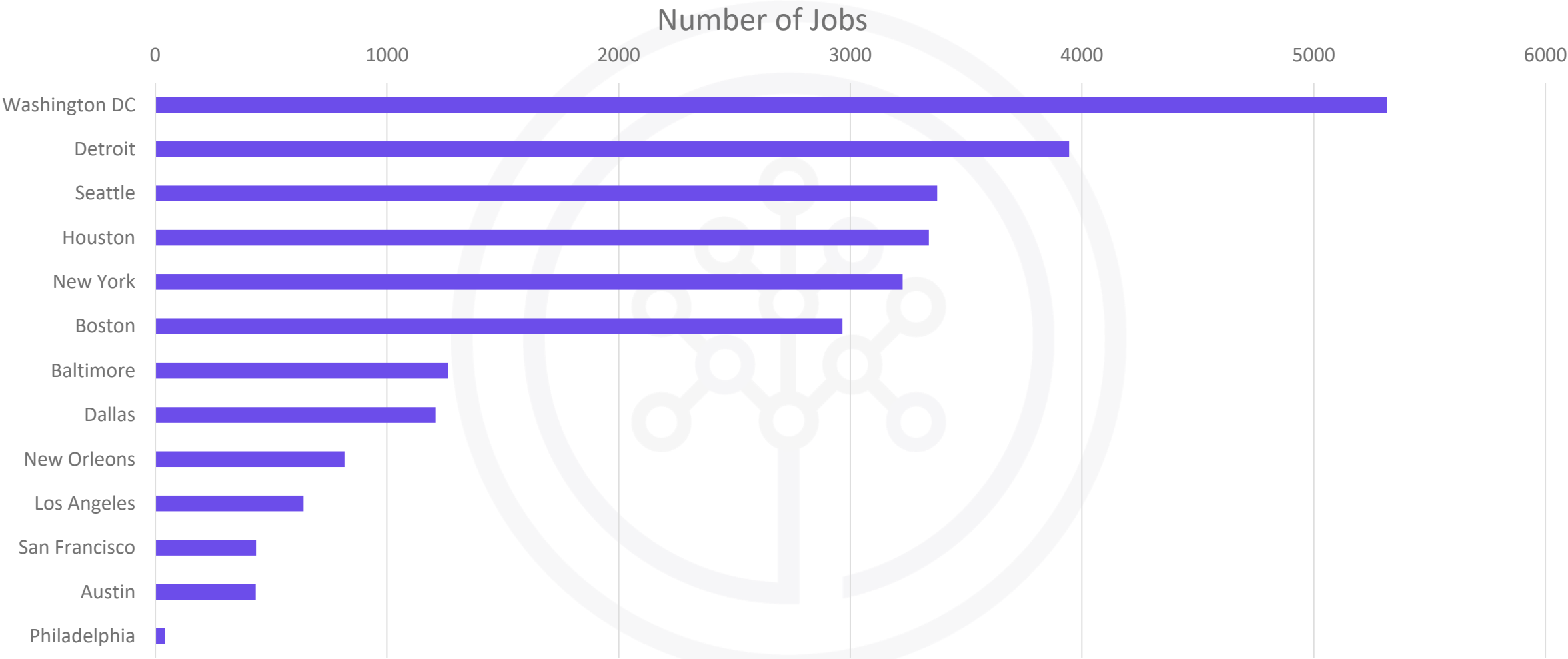
Results



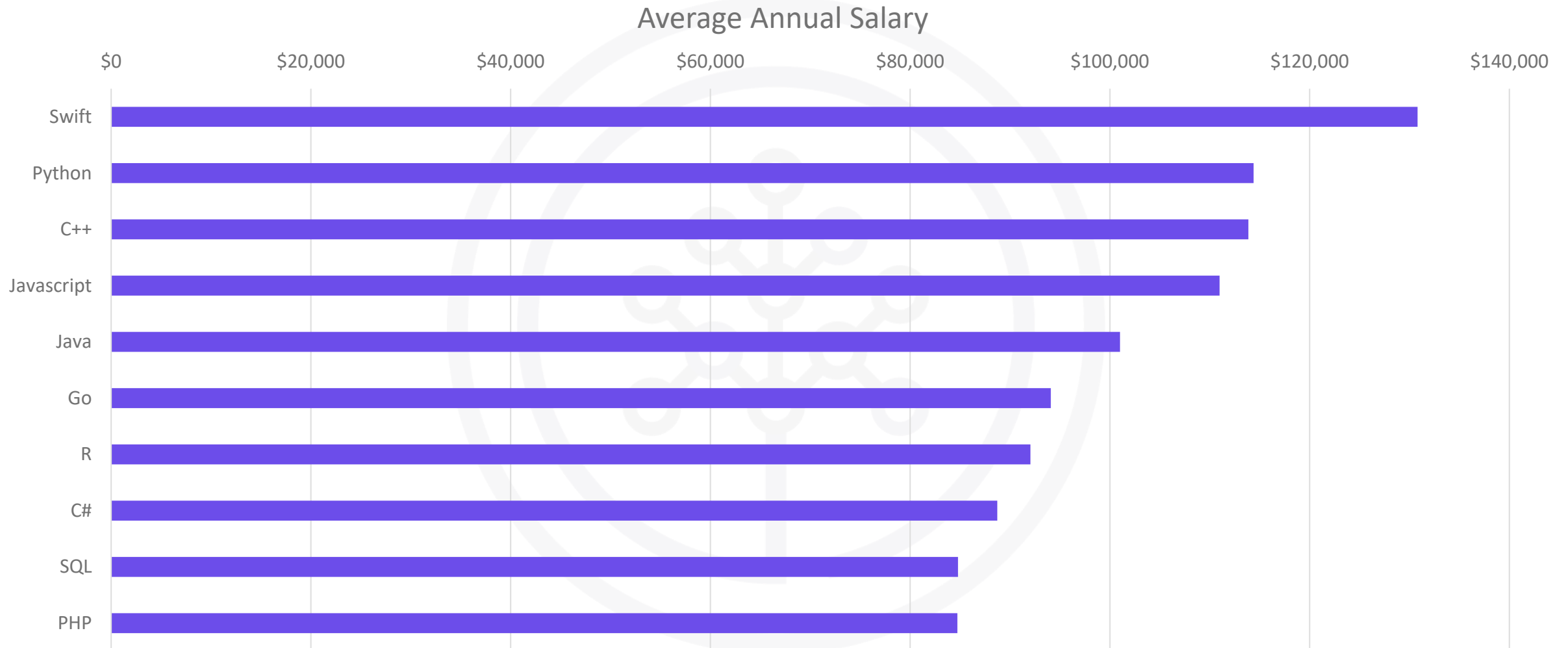
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JOB POSTINGS



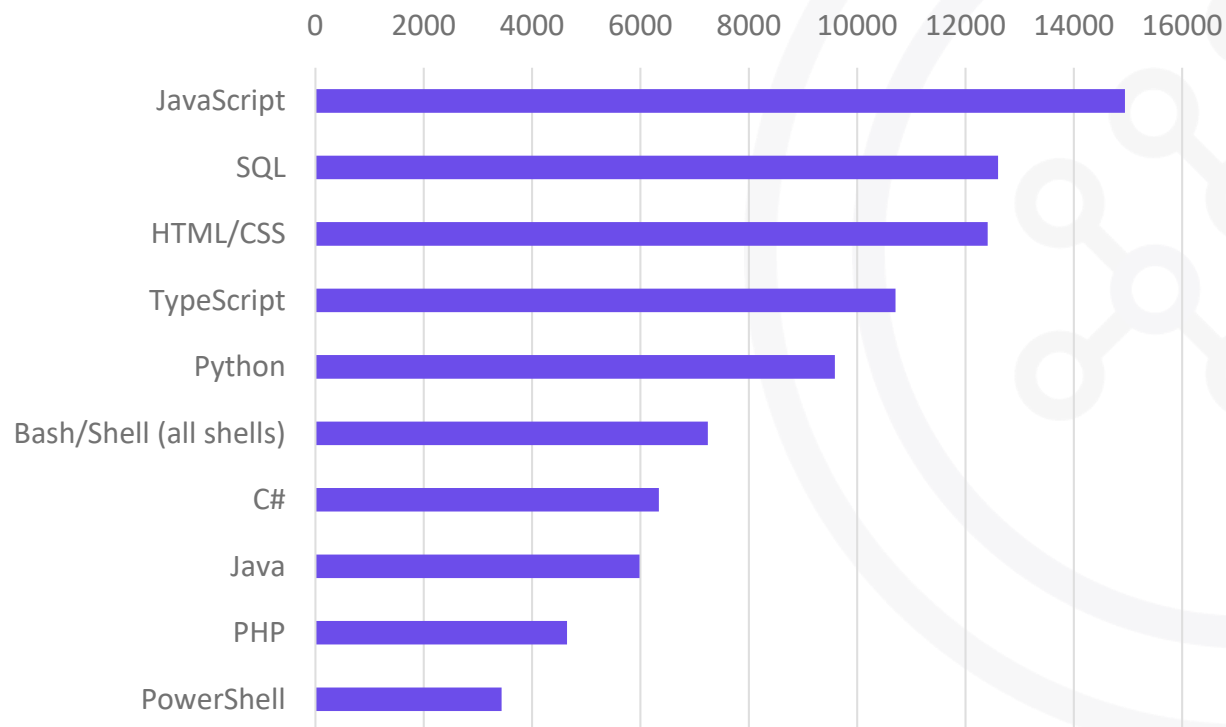
POPULAR LANGUAGES



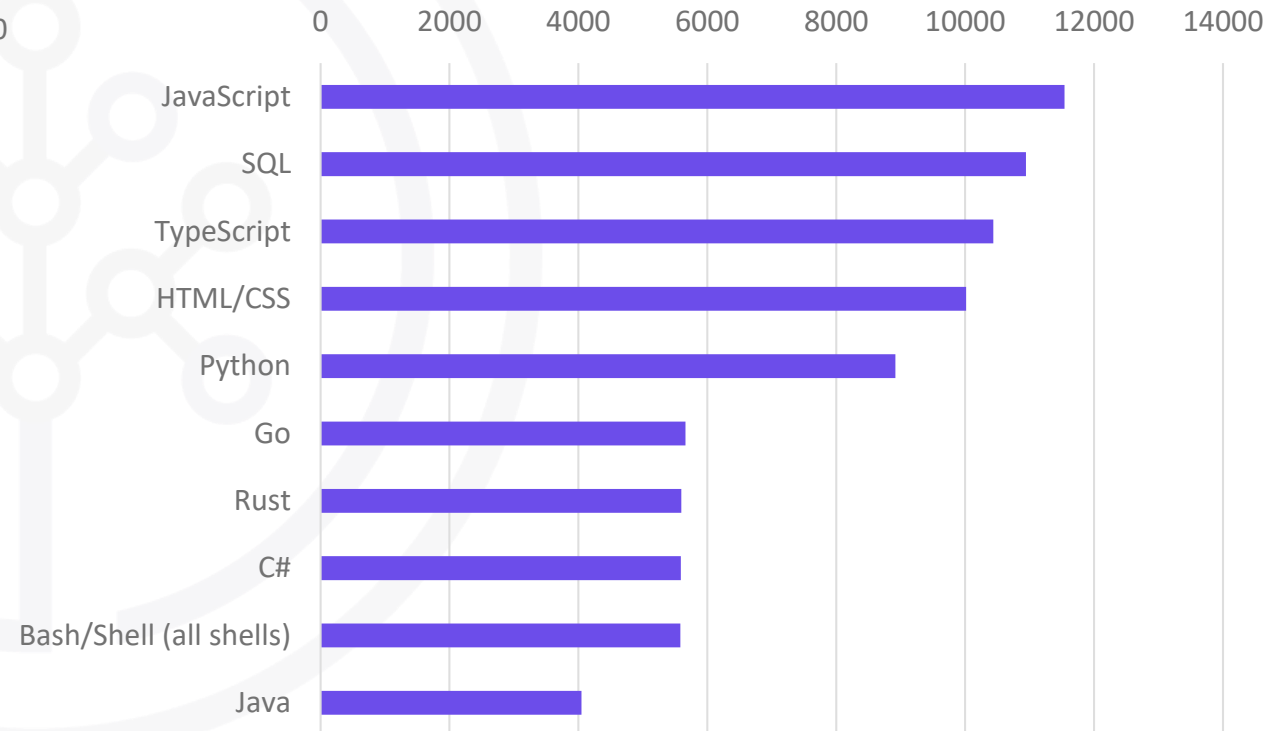
PROGRAMMING LANGUAGE TRENDS

JavaScript remains the most commonly reported programming language in both the current and future periods. SQL and HTML/CSS also remain among the leading technologies, while TypeScript ranks higher in the future outlook.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript remains the most widely reported programming language in both the current and future outlook.
- SQL, HTML/CSS, and Python remain among the leading technologies, indicating continued demand for core web and data-related skills.
- TypeScript gains prominence in the future outlook, while Go and Rust appear in the future Top 10.

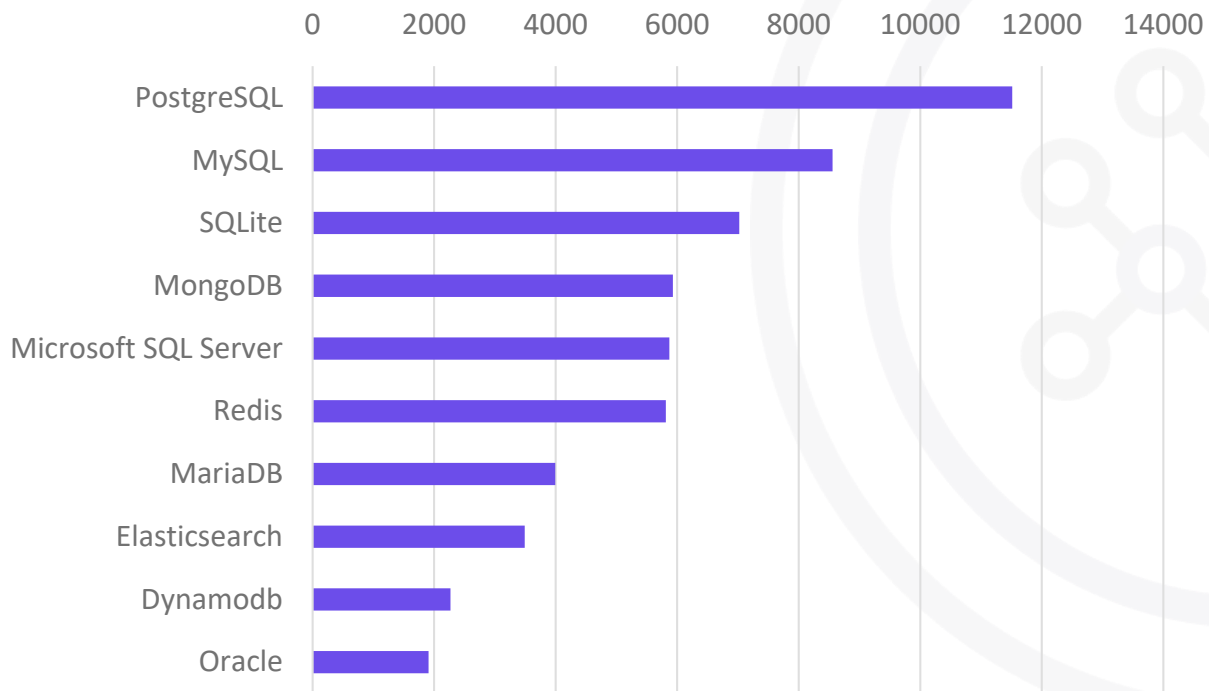
Implications

- Developers should maintain strong JavaScript and SQL skills because they remain consistently prominent.
- Learning TypeScript can help developers prepare for evolving web-development requirements.
- Developers should consider expanding their skills into emerging languages such as Go and Rust to align with future technology trends.

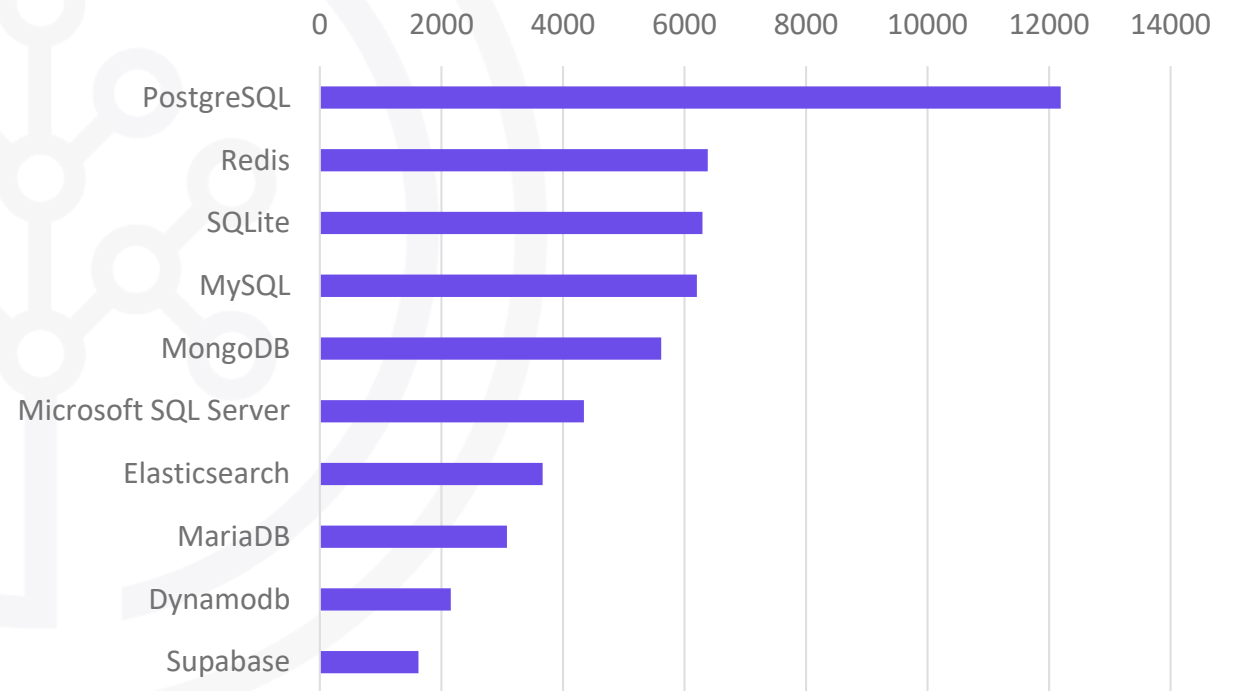
DATABASE TRENDS

PostgreSQL remains the most widely reported database in both the current and future outlook. Redis shows a notable rise in the future ranking, moving from sixth place currently to second place. MySQL remains highly relevant, while newer technologies such as Supabase enter the future Top 10.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL ranks first in both the current and future outlook, indicating consistently strong demand.
- Redis shows the strongest upward movement, rising from 6th place currently to 2nd place in the future outlook.
- Supabase enters the future Top 10, while Oracle drops out, indicating some change in database technology preferences.

Implications

- Developers should maintain strong PostgreSQL skills because it remains the leading database in both periods.
- Learning Redis could be valuable as its position increases significantly in the future outlook.
- Developers should continue monitoring emerging database technologies such as Supabase while maintaining knowledge of established platforms.

DASHBOARD

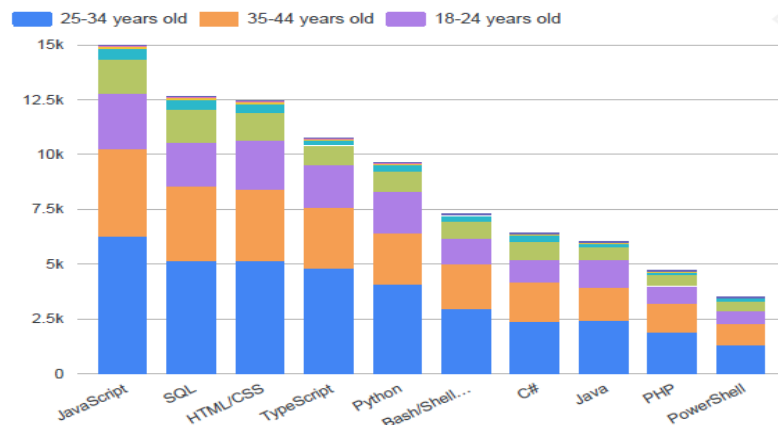


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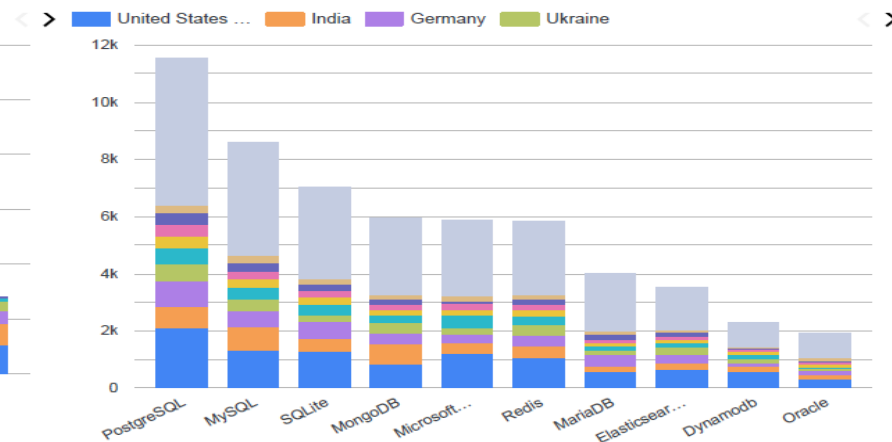


DASHBOARD TAB 1: Current Technology Usage

Top 10 Languages Used



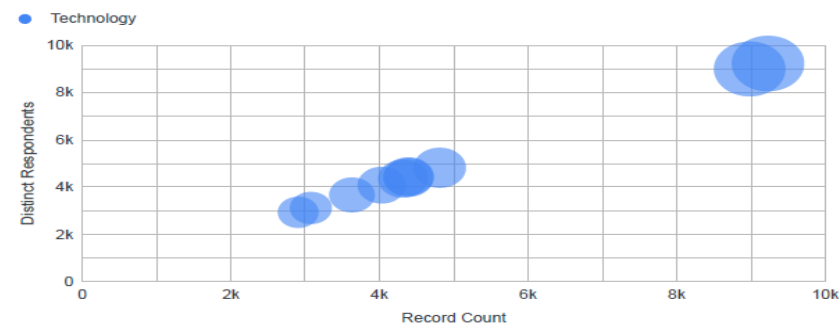
Top 10 Databases Used



Top 10 Platform Used



Top 10 Web Frameworks Used



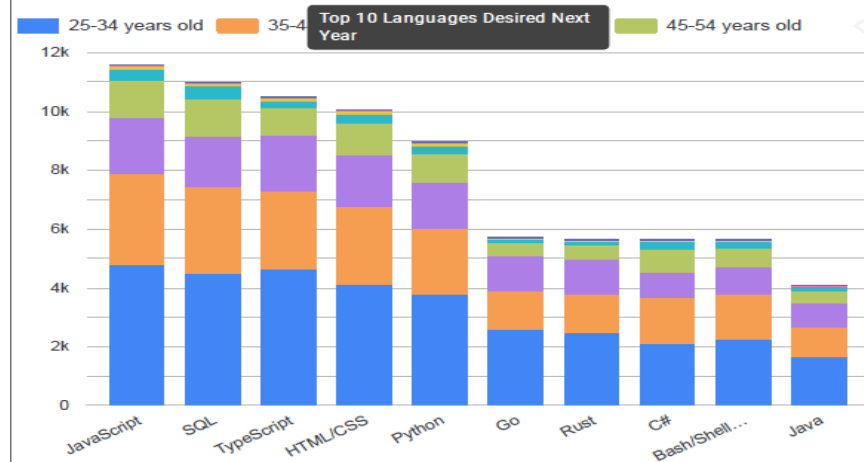
Dashboard Insight

- JavaScript is one of the most widely used programming languages.
- PostgreSQL is the leading database technology in current usage.
- Cloud platforms are prominent in current technology usage, with Microsoft Azure and Google Cloud among the most visible platform
- A small number of web frameworks account for a large share of current usage, with the leading framework clearly ahead of the others.

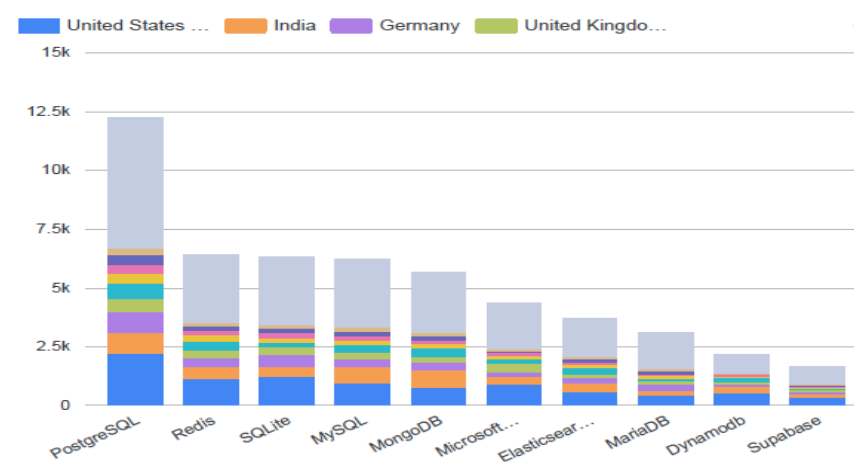


DASHBOARD TAB 2: Future Technology Trends

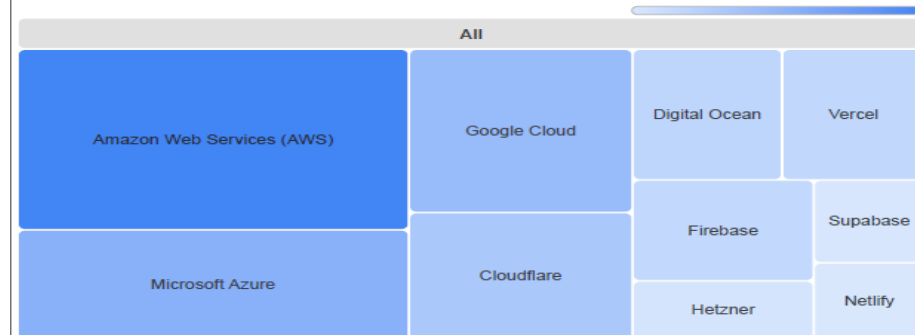
Top 10 Languages Desired Next Year



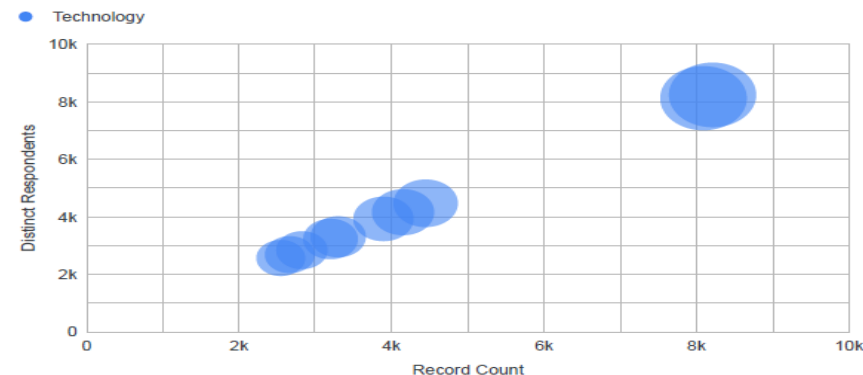
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



Top 10 Desired Web Frameworks



Dashboard Insights

- JavaScript remains the most desired programming language for the next year.
- PostgreSQL remains the most desired database, while Redis rises to second place.
- Amazon Web Services (AWS) is the most prominent desired platform, followed by Microsoft Azure and Google Cloud.
- Web framework demand is concentrated around one leading framework, while the remaining frameworks form a smaller cluster of similar adoption levels.

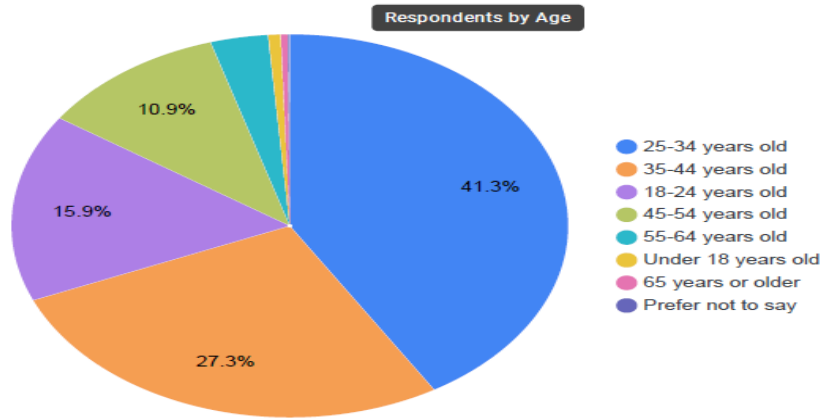


DASHBOARD TAB 3: Demographics

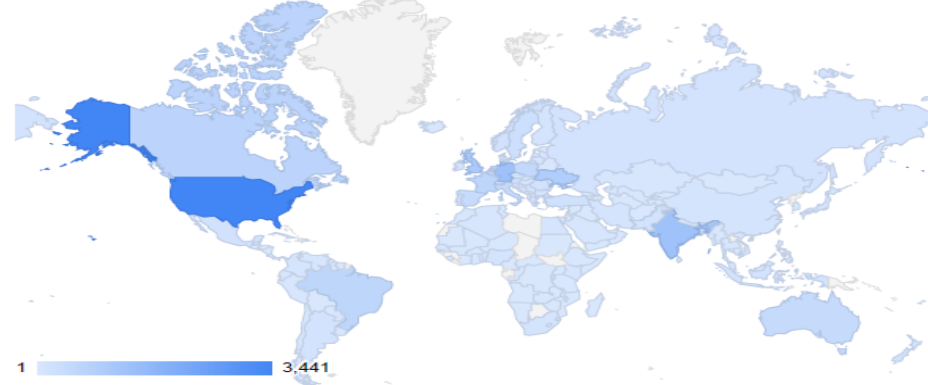
Dashboard Insights

- The respondent population is concentrated in the 25–44 age range, which accounts for 68.6% of respondents.
- Bachelor's degree holders represent the largest education group, followed by Master's degree holders.
- Respondents are geographically distributed across many countries, with the United States showing the highest respondent count.

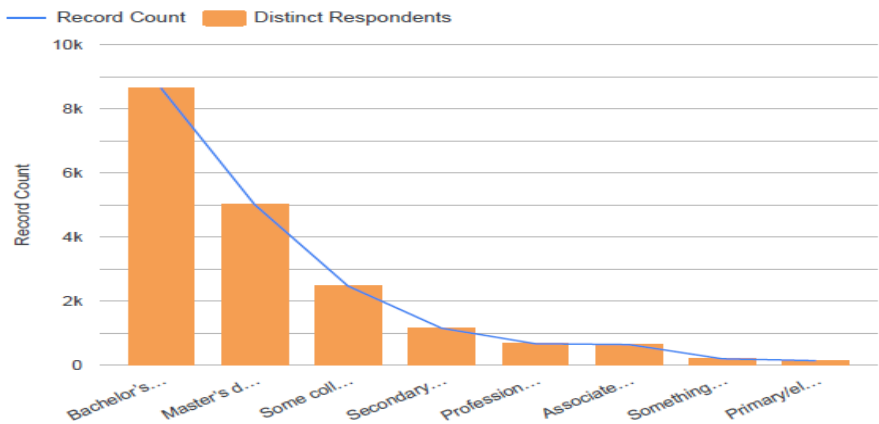
Respondents by Age



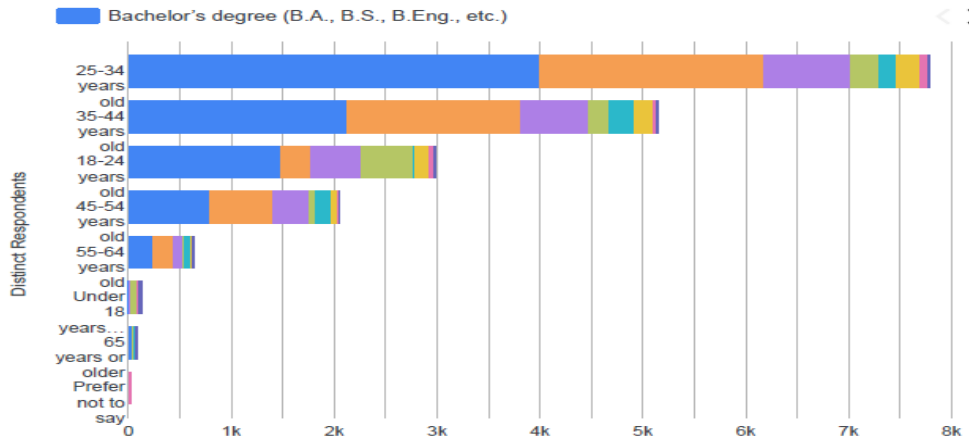
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION

- Discuss overall insights derived from dashboards



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript and PostgreSQL remain leading technologies in both current usage and future preferences.
- Technology preferences show continued demand for established technologies alongside growing interest in technologies such as Redis, TypeScript, Go, Rust, and Supabase.
- The respondent population is concentrated in the 25–44 age range, with Bachelor's degree holders forming the largest education group.

Implications

- Technology professionals should maintain strong skills in established technologies while continuously developing skills in emerging technologies.
- Organizations and training providers should prioritize widely used technologies while incorporating emerging technologies into future skill-development programs.
- The demographic profile indicates that technology-related learning and workforce initiatives can be particularly relevant to professionals in the 25–44 age range.



CONCLUSION



- Programming and database technologies such as JavaScript and PostgreSQL remain highly prominent in current usage and future preferences.
- Future technology preferences show continued demand for established technologies alongside growing interest in technologies such as TypeScript, Redis, Go, Rust, and Supabase.
- Job-posting and salary analysis provides additional insight into technology-related employment opportunities and compensation.
- The dashboards bring together technology usage, future preferences, job-market data, and respondent demographics to support data-driven understanding of the technology landscape.



APPENDIX

